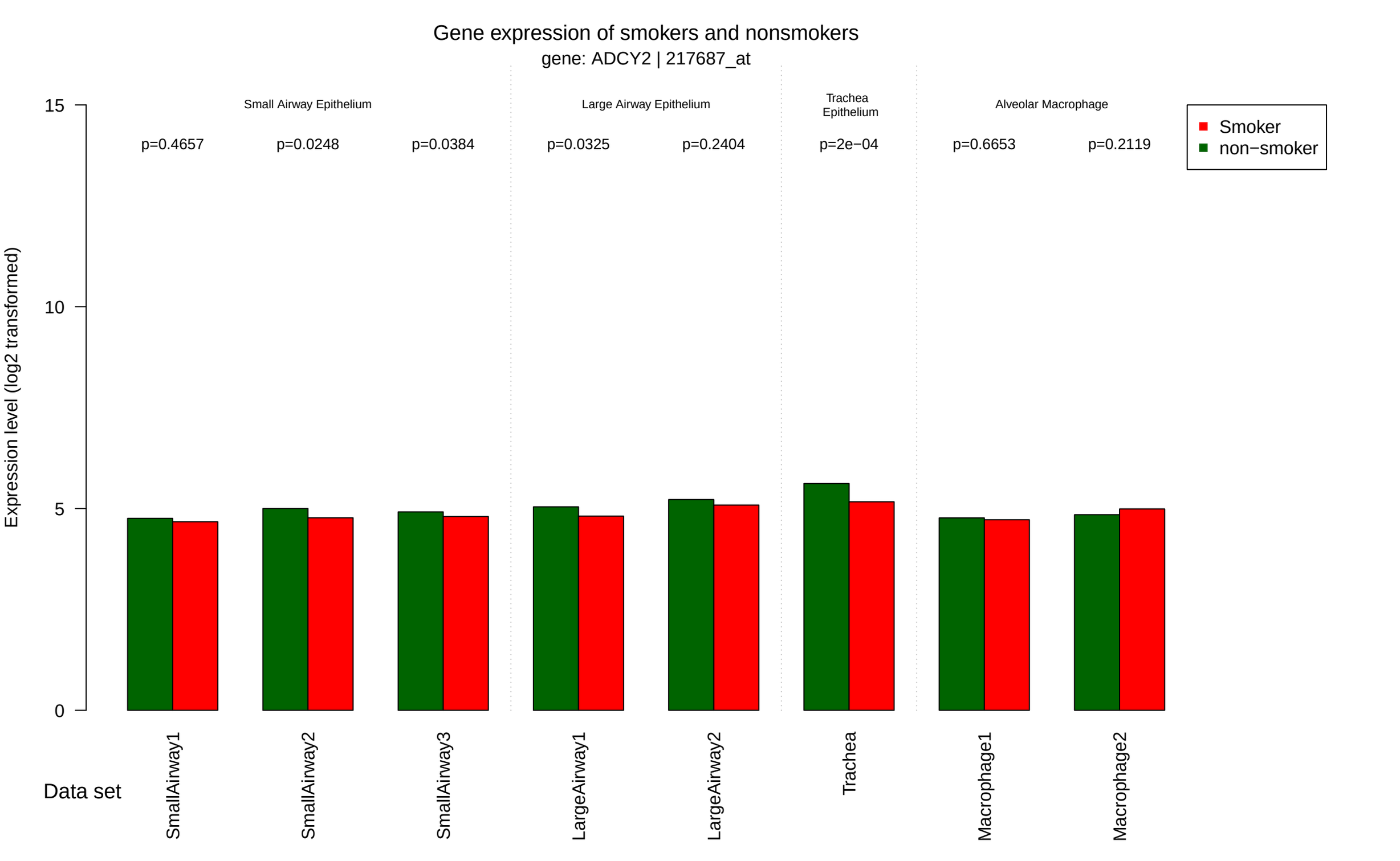
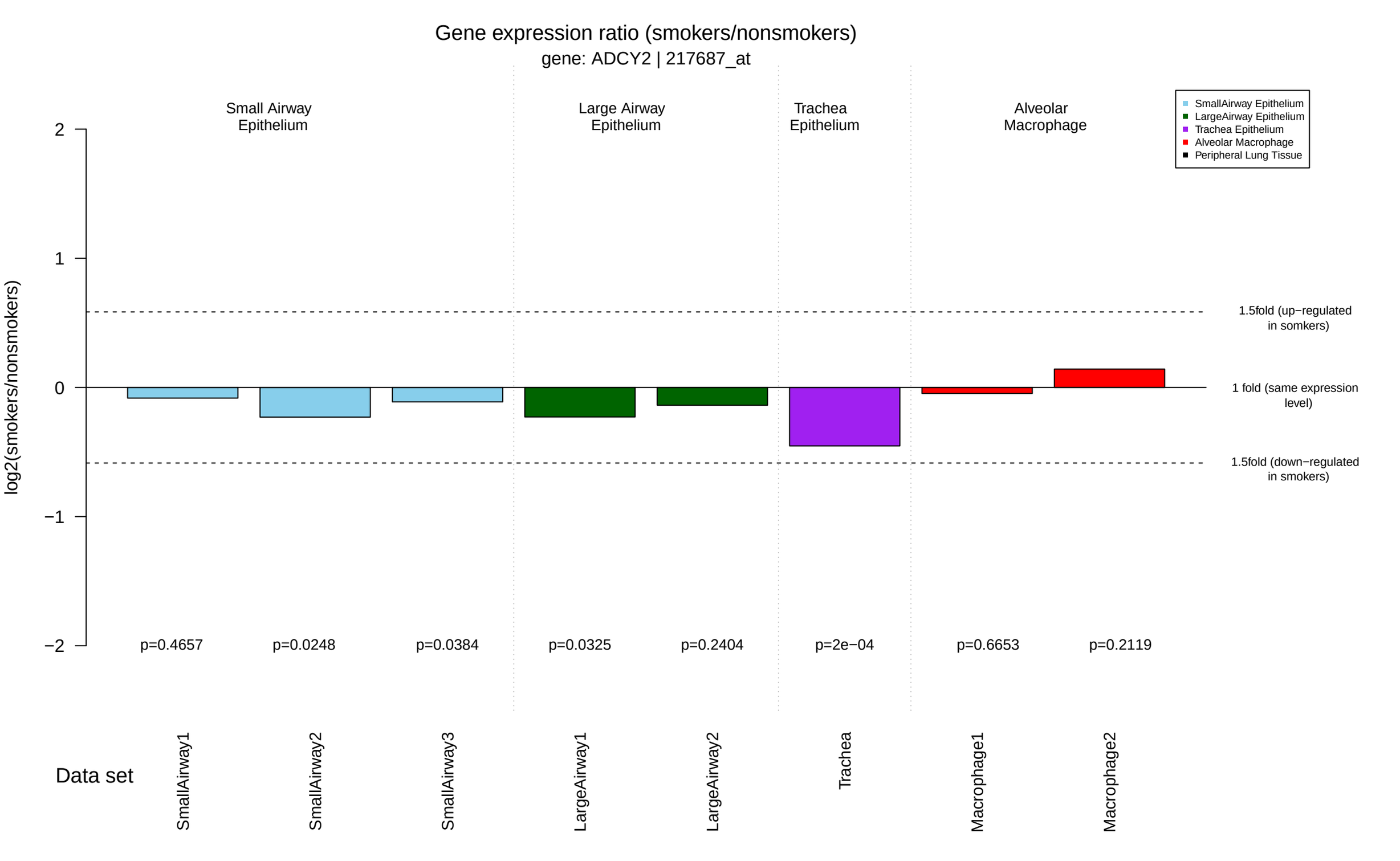
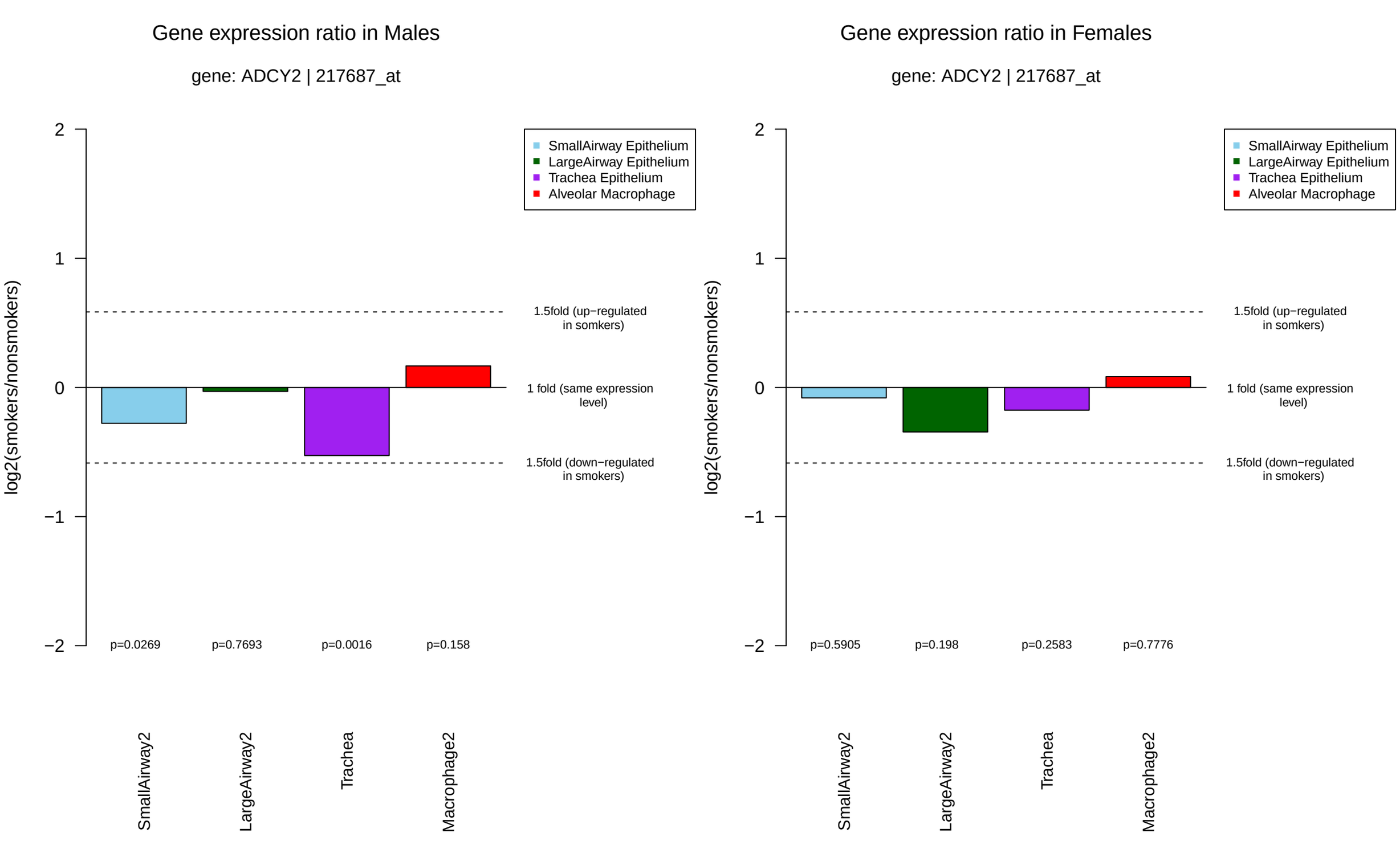
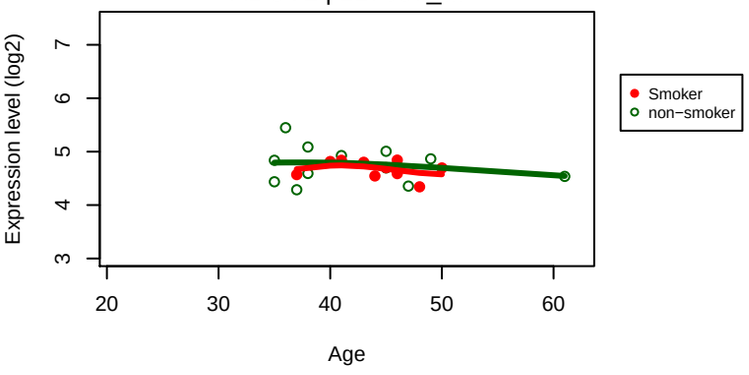
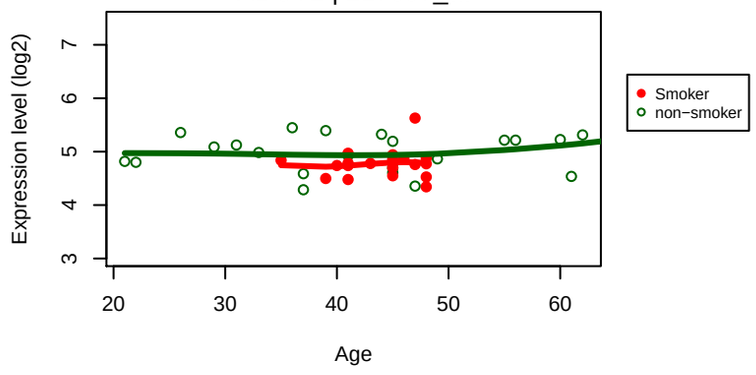
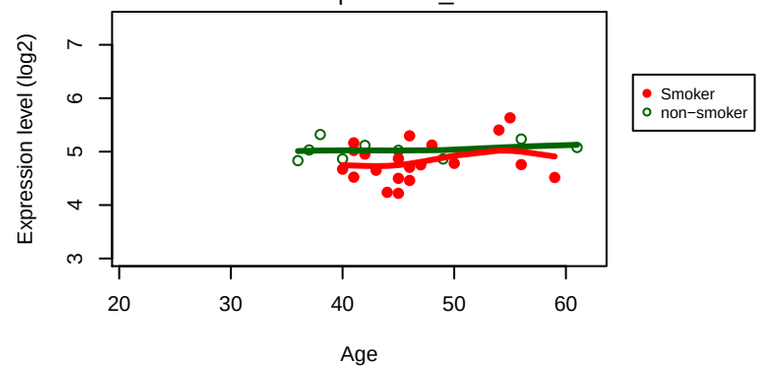

Gene expression of smokers and nonsmokers
gene: ADCY2 | 217687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4657
p=0.0248
p=0.0384
p=0.0325
p=0.2404
p=2e−04
p=0.6653
p=0.2119
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

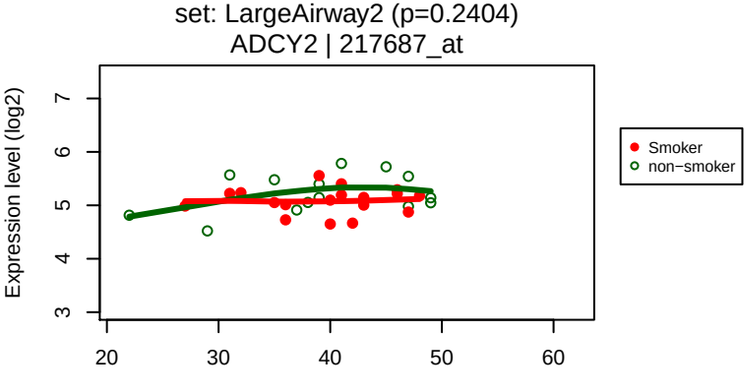

Gene expression ratio (smokers/nonsmokers)
gene: ADCY2 | 217687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4657
p=0.0248
p=0.0384
p=0.0325
p=0.2404
p=2e−04
p=0.6653
p=0.2119
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

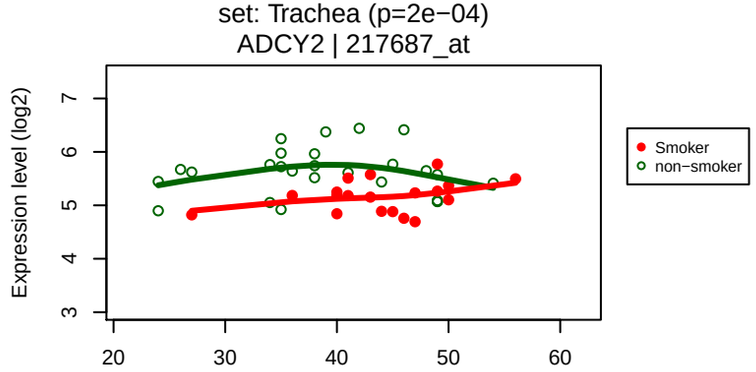

Gene expression ratio in Males
gene: ADCY2 | 217687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0269
p=0.7693
p=0.0016
p=0.158
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY2 | 217687_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5905
p=0.198
p=0.2583
p=0.7776
SmallAirway2
LargeAirway2
Trachea
Macrophage2

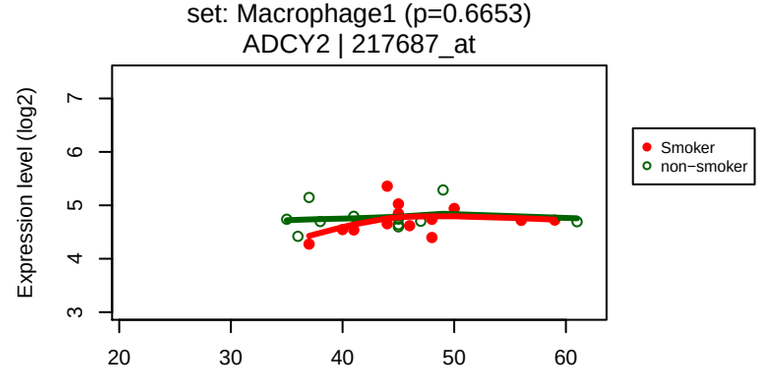

set: SmallAirway1 (p=0.4657)
ADCY2 | 217687_at
Expression level (log2)
Age
Smoker
non−smoker

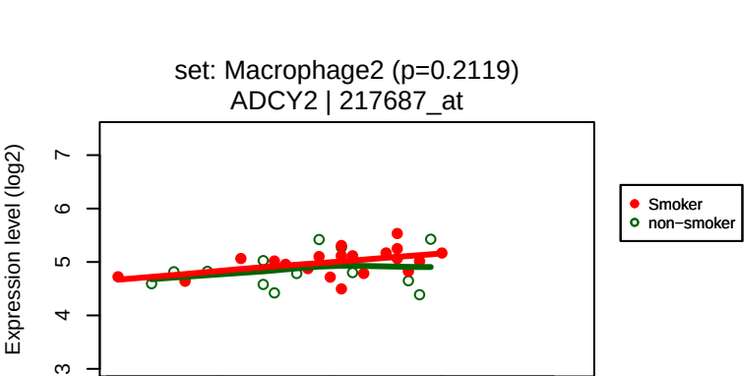

set: SmallAirway2 (p=0.0248)
ADCY2 | 217687_at
Expression level (log2)
Age
Smoker
non−smoker

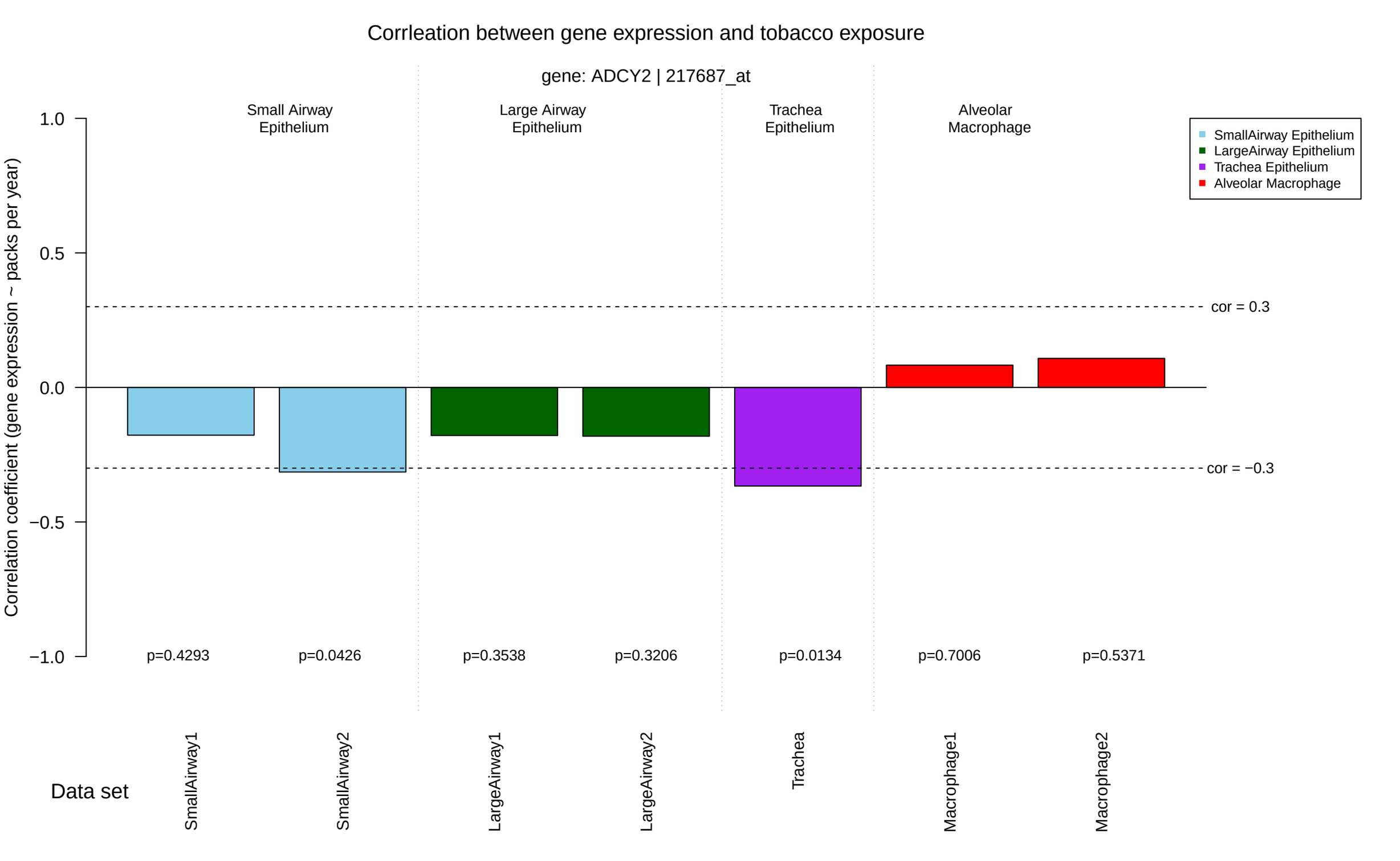

set: LargeAirway1 (p=0.0325)
ADCY2 | 217687_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2404)
ADCY2 | 217687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
ADCY2 | 217687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6653)
ADCY2 | 217687_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2119)
ADCY2 | 217687_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY2 | 217687_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4293
p=0.0426
p=0.3538
p=0.3206
p=0.0134
p=0.7006
p=0.5371
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2